Prompting

- Prompting can be interpreted as a fine-tuning alternative (François Chollet)
- TELER, Taxonomy for prompting LLMs, Source
- Prompt ressources : Link

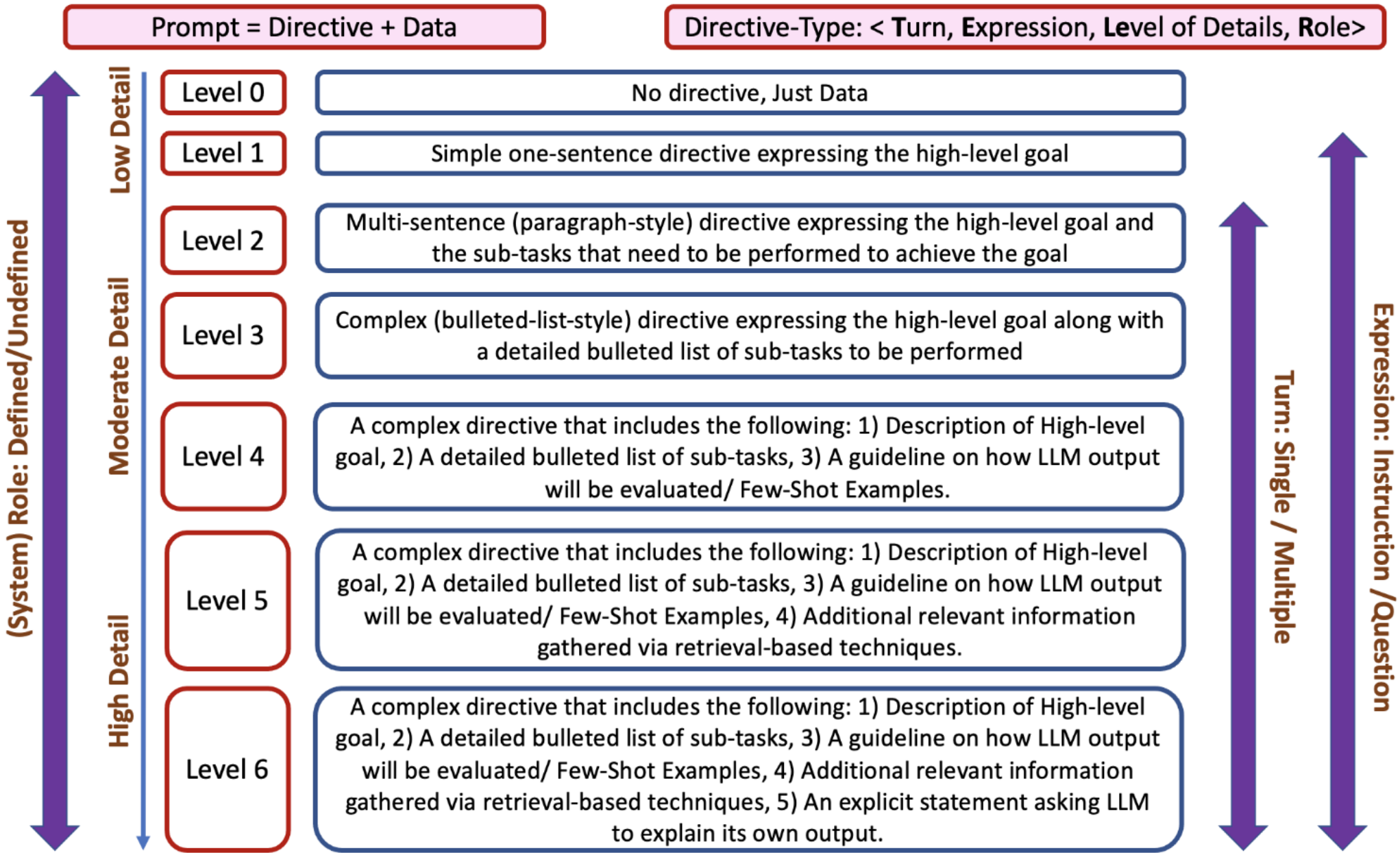


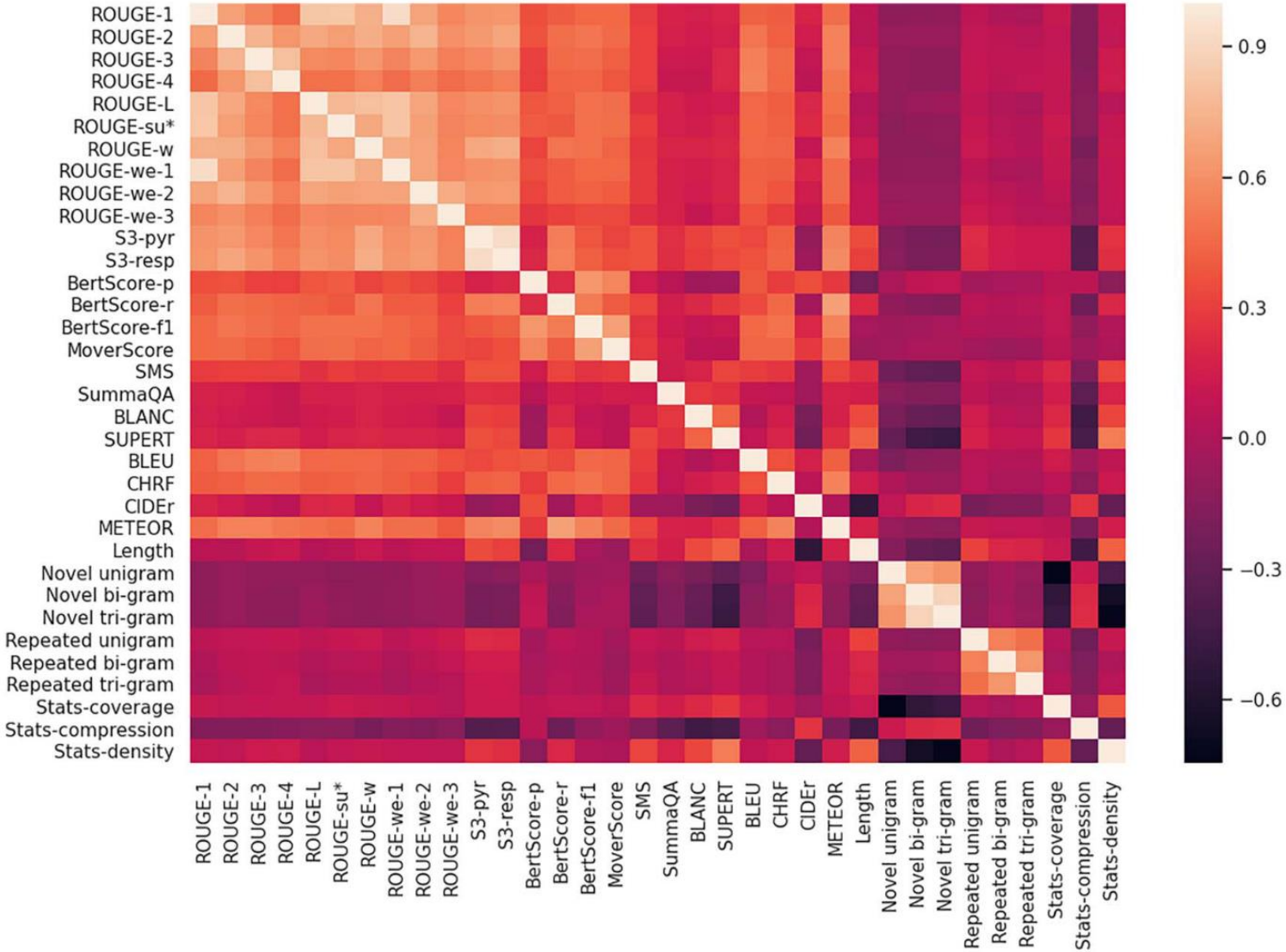
Figure 1: Proposed Prompt Taxonomy: **TELeR** (<**T**urn, **E**xpression, **Le**vel of Details, **R**ole>)

Evaluation (the darkside of NLP)

- Case study :

Text Summarization

Source



Using human annotators

Instructions

In this task you will evaluate the quality of summaries written for a news article.
To correctly solve this task, follow these steps:

1. Carefully read the news article, be aware of the information it contains.
2. Read the proposed summaries A-F (6 in total).
3. Rate each summary on a scale from **1** (worst) to **5** (best) by its *relevance*, *consistency*, *fluency*, and *coherence*.

Definitions

Relevance:

The rating measures how well the summary captures the key points of the article.
Consider whether all and only the important aspects are contained in the summary.

Consistency:

The rating measures whether the facts in the summary are consistent with the facts in the original article.
Consider whether the summary does reproduce all facts accurately and does not make up untrue information.

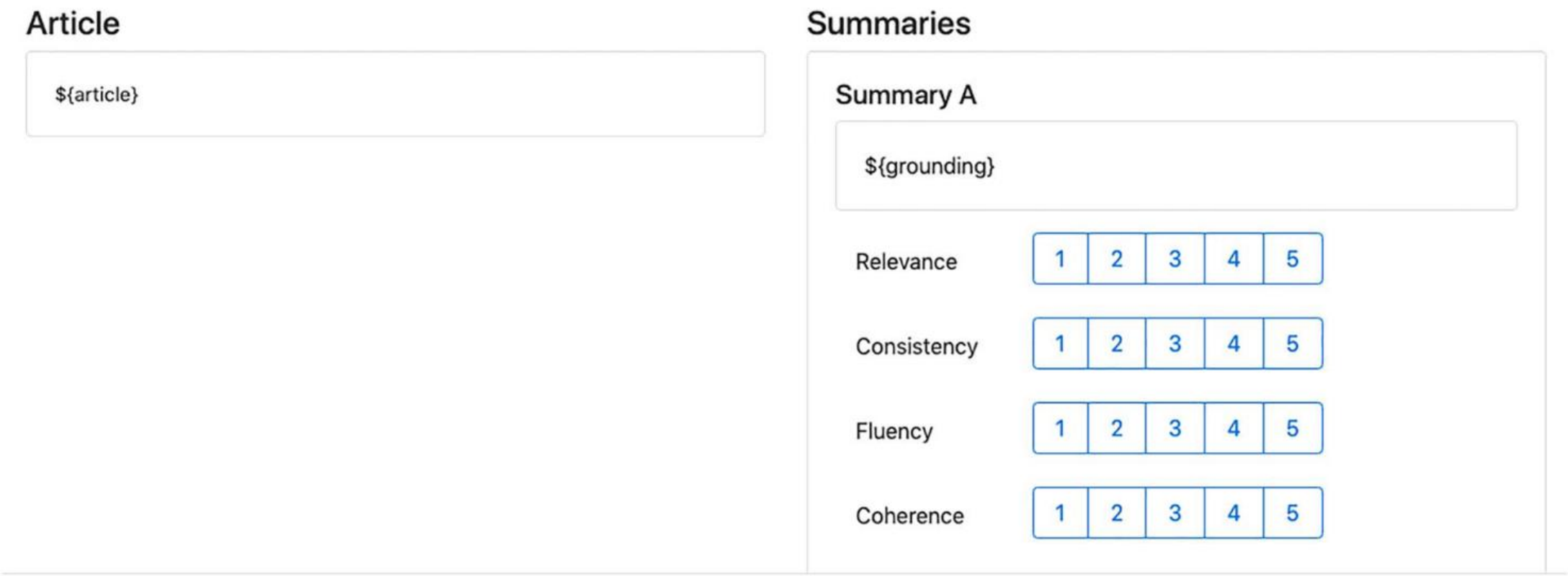
Fluency

This rating measures the quality of individual sentences, are they well-written and grammatically correct.
Consider the quality of individual sentences.

Coherence:

The rating measures the quality of all sentences collectively, to the fit together and sound naturally.
Consider the quality of the summary as a whole.

Annotator interface



Kendall's tau correlation coefficients of expert annotations computed on a system-level along four quality dimensions with automatic metrics using 11 reference summaries per example. ˆ denotes metrics which use the source document. The five most-correlated metrics in each column are bolded

Metric	Coherence	Consistency	Fluency	Relevance
ROUGE-1	0.2500	0.5294	**0.5240**	0.4118
ROUGE-2	0.1618	0.5882	0.4797	0.2941
ROUGE-3	0.2206	**0.7059**	**0.5092**	0.3529
ROUGE-4	**0.3088**	0.5882	**0.5535**	0.4118
ROUGE-L	0.0735	0.1471	0.2583	0.2353
ROUGE-su*	0.1912	0.2941	0.4354	0.3235
ROUGE-w	0.0000	0.3971	0.3764	0.1618
ROUGE-we-1	**0.2647**	0.4559	**0.5092**	**0.4265**
ROUGE-we-2	−0.0147	0.5000	0.3026	0.1176
ROUGE-we-3	0.0294	0.3676	0.3026	0.1912
S^3-pyr	−0.0294	0.5147	0.3173	0.1324
S^3-resp	−0.0147	0.5000	0.3321	0.1471
BertScore-p	0.0588	−0.1912	0.0074	0.1618
BertScore-r	0.1471	**0.6618**	0.4945	0.3088
BertScore-f	0.2059	0.0441	0.2435	**0.4265**
MoverScore	0.1912	−0.0294	0.2583	0.2941
SMS	0.1618	0.5588	0.3616	0.2353
SummaQAˆ	0.1176	**0.6029**	0.4059	0.2206
BLANCˆ	0.0735	0.5588	0.3616	0.2647
SUPERTˆ	0.1029	0.5882	0.4207	0.2353
BLEU	0.1176	0.0735	0.3321	0.2206
CHRF	**0.3971**	0.5294	0.4649	**0.5882**
CIDEr	0.1176	−0.1912	−0.0221	0.1912
METEOR	0.2353	**0.6324**	**0.6126**	**0.4265**
Lengthˆ	−0.0294	0.4265	0.2583	0.1618
Novel unigramˆ	0.1471	−0.2206	−0.1402	0.1029
Novel bi-gramˆ	0.0294	−0.5441	−0.3469	−0.1029
Novel tri-gramˆ	0.0294	−0.5735	−0.3469	−0.1324
Repeated unigramˆ	**−0.3824**	0.1029	−0.0664	−0.3676
Repeated bi-gramˆ	**−0.3824**	−0.0147	−0.2435	**−0.4559**
Repeated tri-gramˆ	−0.2206	0.1471	−0.0221	−0.2647
Stats-coverageˆ	−0.1324	0.3529	0.1550	−0.0294
Stats-compressionˆ	0.1176	−0.4265	−0.2288	−0.0147
Stats-densityˆ	0.1618	**0.6471**	0.3911	0.2941

Human ratings of summaries along four evaluation dimensions, averaged over three expert annotators, broken down by extractive and abstractive models. The M* codes follow the notation described in Section 3.2. The three highest-rated models in each column are in bold.

Method	Coherence	Consistency	Fluency	Relevance
CNN/DM Reference Summary	3.26	4.47	4.79	3.77
Extractive Models				
M0 - **LEAD-3**	**4.16**	4.98	4.94	4.14
M1 - **NEUSUM**	3.22	4.98	4.90	3.82
M2 - **BanditSum**	3.28	4.99	4.83	3.81
M5 - **RNES**	3.71	**4.97**	4.81	4.06
Abstractive Models				
M8 - **Pointer Generator**	3.29	4.65	4.79	3.55
M9 - **Fast-abs-rl**	2.38	4.67	4.50	3.52
M10 - **Bottom-Up**	2.73	4.25	4.42	3.38
M11 - **Improve-abs**	2.28	3.27	3.65	3.15
M12 - **Unified-ext-abs**	3.60	**4.96**	4.85	3.85
M13 - **ROUGESal**	3.44	4.82	4.86	3.83
M14 - **Multi-task (Ent + QG)**	3.20	4.90	4.74	3.63
M15 - **Closed book decoder**	3.35	**4.95**	4.80	3.67
M17 - **T5**	4.00	4.93	**4.93**	4.23
M20 - **GPT-2** (zero shot)[1]	3.63	3.40	3.97	3.30
M22 - **BART**	**4.18**	4.94	**4.90**	**4.25**
M23 - **Pegasus** (C4)	**4.16**	4.91	**4.88**	**4.26**
M23 - **Pegasus** (dynamic mix)	4.09	4.85	4.79	**4.27**

Method	ROUGE-1/2/3/4/L/su*/w	ROUGE-WE-(1/2/3)	S^3 (pyr/resp)	BertScore	MoverScore	SummaQA	SMS	BLANC	SUPERT
			Extractive Models						
M0 - **LEAD-3**	0.3994 / 0.1746 / 0.0990 / 0.0647 / 0.3606 / 0.1377 / 0.2072	0.4049 / 0.2260 / 0.2172	**0.5395 / 0.6328**	0.3742	0.1679	**0.1652**	0.1050	0.0480	**0.7259**
M1 - **NEUSUM**	0.4130 / 0.1893 / 0.1109 / 0.0742 / 0.3768 / 0.1495 / 0.2156	0.4186 / 0.2402 / 0.2310	**0.5562 / 0.6509**	0.3955	0.1839	**0.1700**	**0.1062**	**0.1087**	0.7010
M2 - **BanditSum**	0.4137 / 0.1868 / 0.1086 / 0.0721 / 0.3759 / 0.1513 / 0.2139	0.4195 / 0.2385 / 0.2300	**0.5339** / 0.6306	0.3938	0.1815	0.1324	**0.1058**	0.0909	**0.7018**
M3 - **LATENT**	0.4136 / 0.1867 / 0.1085 / 0.0721 / 0.3757 / 0.1512 / 0.2138	0.4194 / 0.2384 / 0.2299	0.5337 / 0.6305	0.3936	0.1814	**0.1645**	0.1058	0.0910	**0.7020**
M4 - **REFRESH**	0.3972 / 0.1807 / 0.1042 / 0.0690 / 0.3621 / 0.1340 / 0.2129	0.4023 / 0.2318 / 0.2238	**0.6395 / 0.7124**	0.3903	0.1720	**0.1944**	**0.1088**	**0.1406**	**0.7526**
M5 - **RNES**	0.4088 / 0.1878 / 0.1102 / 0.0736 / 0.3719 / 0.1446/ **0.2163**	0.4153 / 0.2395 / 0.2317	**0.6082 / 0.6894**	**0.3997**	0.1802	**0.1794**	**0.1107**	**0.1232**	**0.7434**
M6 - **JECS**	0.4144 / 0.1846 / 0.1063 / 0.0699 / 0.3760 / 0.1485 / 0.2135	0.4200 / 0.2371 / 0.2283	0.5337 / 0.6284	0.3925	0.1805	0.1644	0.1048	**0.1044**	0.6946
M7 - **STRASS**	0.3377 / 0.1237 / 0.0650 / 0.0416 / 0.2790 / 0.1052 / 0.1559	0.3477 / 0.1757 / 0.1656	0.3632 / 0.4939	0.3090	0.1079	0.1367	0.1023	**0.1042**	0.6566
			Abstractive Models						
M8 - **Pointer Generator**	0.3921 / 0.1723 / 0.1003 / 0.0674 / 0.3599 / 0.1435 / 0.1999	0.3990 / 0.2226 / 0.2128	0.4328 / 0.5561	0.3763	0.1643	0.1398	0.0974	0.0704	0.6501
M9 - **Fast-abs-rl**	0.4057 / 0.1774 / 0.0975 / 0.0616 / 0.3806 / 0.1439 / 0.2112	0.4123 / 0.2302 / 0.2184	0.4818 / 0.5865	0.3918	0.1748	0.1431	0.0847	0.0855	0.6125
M10 - **Bottom-Up**	0.4124 / 0.1870 / 0.1064 / 0.0695 / **0.3815** / 0.1543 / 0.2084	0.4192 / 0.2400 / 0.2313	0.4450 / 0.5655	0.3964	0.1830	0.1408	0.0925	0.0570	0.6092
M11 - **Improve-abs**	0.3985 / 0.1720 / 0.0927 / 0.0567 / 0.3730 / 0.1431 / 0.2073	0.4045 / 0.2300 / 0.2228	0.4899 / 0.5897	0.3826	0.1652	0.1341	0.0816	0.0777	0.5972
M12 - **Unified-ext-abs**	0.4038 / 0.1790 / 0.1039 / 0.0695 / 0.3675 / 0.1484 / 0.2074	0.4097 / 0.2299 / 0.2204	0.4936 / 0.5995	0.3832	0.1739	0.1530	0.1038	0.0962	0.6826
M13 - **ROUGESal**	0.4016 / 0.1797 / 0.1053 / 0.0709 / 0.3679 / 0.1497 / 0.2058	0.4078 / 0.2294 / 0.2190	0.4643 / 0.5799	0.3837	0.1722	0.1475	0.1009	0.0882	0.6570
M14 - **Multi-task (Ent + QG)**	0.3952 / 0.1758 / 0.1037 / 0.0705 / 0.3625 / 0.1476 / 0.2007	0.4015 / 0.2253 / 0.2149	0.4246 / 0.5513	0.3759	0.1670	0.1360	0.0982	0.0648	0.6380
M15 - **Closed book decoder**	0.3976 / 0.1760 / 0.1031 / 0.0696 / 0.3636 / 0.1472 / 0.2033	0.4039 / 0.2263 / 0.2160	0.4591 / 0.5757	0.3783	0.1699	0.1456	0.1009	0.0896	0.6612
M16 - **SENECA**	0.4151 / 0.1836 / 0.1052 / 0.0681 / 0.3806 / 0.1520 / 0.2112	0.4211 / 0.2369 / 0.2282	0.4735 / 0.5836	0.3907	0.1811	0.1404	0.1005	0.0692	0.6519
M17 - **T5**	**0.4479 / 0.2205 / 0.1336 / 0.0920 / 0.4172 / 0.1879 / 0.2291**	**0.4543 / 0.2723 / 0.2631**	0.5168 / 0.6294	**0.4450**	**0.2376**	0.1437	0.1046	0.0773	0.6094
M18 - **NeuralTD**	0.4004 / 0.1762 / 0.1000 / 0.0650 / 0.3723 / 0.1452 / 0.2085	0.4063 / 0.2277 / 0.2187	0.4946 / 0.5975	0.3949	0.1697	0.1440	0.0916	0.0859	0.6290
M19 - **BertSum-abs**	**0.4163 / 0.1944 / 0.1156 / 0.0785** / 0.3554 / **0.1625** / 0.1979	**0.4230 / 0.2454 / 0.2351**	0.4664 / 0.5855	0.3855	**0.1894**	0.1385	**0.1071**	0.0815	0.6116
M20 - **GPT-2** (supervised)	0.3981 / 0.1758 / 0.0993 / 0.0649 / 0.3674 / 0.1470 / 0.2006	0.4048 / 0.2268 / 0.2170	0.4069 / 0.5373	0.3915	0.1750	0.1299	0.0930	0.0705	0.6053
M21 - **UniLM**	**0.4306 / 0.2044 / 0.1218 / 0.0824 / 0.4013 / 0.1714 / 0.2228**	**0.4369 / 0.2567 / 0.2483**	0.5143 / 0.6210	**0.4122**	**0.2112**	0.1455	0.0957	0.0841	0.6100
M22 - **BART**	**0.4416 / 0.2128 / 0.1285 / 0.0880 / 0.4100 / 0.1818 / 0.2266**	**0.4472 / 0.2646 / 0.2556**	0.5116 / 0.6215	**0.4264**	**0.2259**	0.1457	0.1037	0.0822	0.6184
M23 - **Pegasus** (dynamic mix)	**0.4407 / 0.2155 / 0.1307 / 0.0901 / 0.4101 / 0.1825 / 0.2260**	**0.4471 / 0.2668 / 0.2575**	0.5099 / **0.6233**	**0.4369**	**0.2283**	0.1422	0.1040	0.0797	0.6046
M23 - **Pegasus** (huge news)	**0.4408 / 0.2147 / 0.1295 / 0.0889 / 0.4103 / 0.1821 / 0.2273**	**0.4473 / 0.2663 / 0.2568**	0.5295 / **0.6372**	**0.4377**	**0.2286**	0.1497	0.1049	0.0845	0.6148

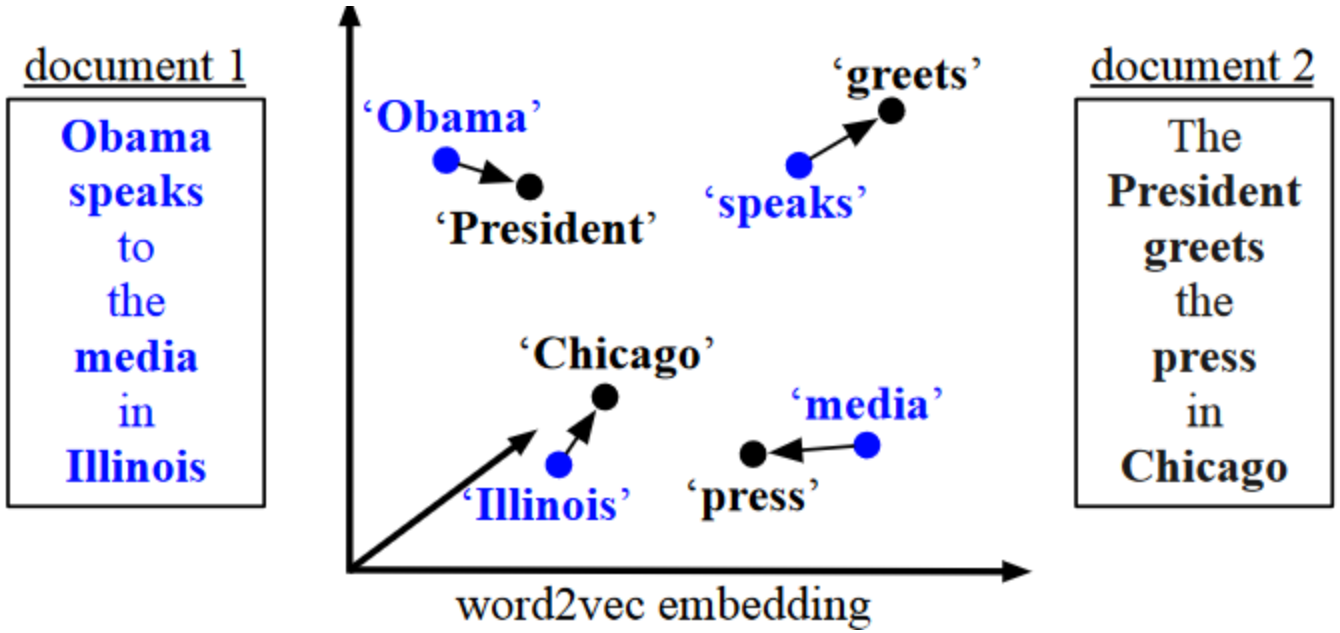
Method	BLEU	CHRF	CIDEr	METEOR	Length	Stats (cov/comp/den)	Repeated (1/2/3)
	Extractive Models						
M0 - **LEAD-3**	11.4270	0.3892	0.2125	**0.2141**	87.4475	0.9825 / 9.6262 / 57.8001	0.2086 / 0.0310 / 0.0310
M1 - textbfNEUSUM	12.7784	0.3946	0.2832	**0.2183**	84.4075	0.9819 / 9.8047 / 32.8574	0.2325 / 0.0531 / 0.0531
M2 - **BanditSum**	12.9761	0.3897	0.3305	0.2124	78.5279	0.9836 / 10.2810 / 40.4265	0.2384 / 0.0573 / 0.0573
M3 - **LATENT**	12.9725	0.3897	0.3305	0.2123	78.5279	0.9834 / 10.2809 / 40.4095	0.2384 / 0.0573 / 0.0573
M4 - **REFRESH**	10.6568	**0.4526**	0.0677	**0.2395**	114.5684	0.9850 / 7.1059 / 53.1928	0.2127 / 0.0289 / 0.0289
M5 - **RNES**	11.2203	**0.4062**	0.1559	**0.2300**	99.9199	0.9938 / 7.9032 / 67.7089	0.2451 / 0.0540 / 0.0540
M6 - **JECS**	12.5659	**0.4310**	0.3090	0.2122	79.7797	0.9874 / 10.1111 / 26.6943	0.2041 / 0.0327 / 0.0327
M7 - **STRASS**	7.8330	0.3330	0.2945	0.1607	76.4859	0.9969 / 12.7835 / 59.9498	0.1864 / 0.0343 / 0.0343
	Abstractive Models						
M8 - **Pointer Generator**	13.8247	0.3567	0.5065	0.1860	63.5211	0.9957 / 13.1940 / 26.0880	0.2015 / 0.0375 / 0.0375
M9 - **Fast-abs-rl**	12.9812	0.3778	0.4329	0.2014	70.8600	0.9860 / 11.0141 / **9.9859**	0.2157 / 0.0370 / 0.0370
M10 - **Bottom-Up**	**15.1293**	0.3523	**0.6176**	0.1887	**56.5715**	0.9811 / **14.7771** / 12.6181	**0.1856 / 0.0211 / 0.0211**
M11 - **Improve-abs**	11.9816	0.3715	0.3356	0.2005	75.9512	**0.9674** / 10.6043 / **8.9755**	0.2499 / 0.0542 / 0.0542
M12 - **Unified-ext-abs**	12.8457	0.3786	0.3851	0.2017	74.4663	0.9868 / 10.7510 / 33.1106	0.2177 / 0.0493 / 0.0493
M13 - **ROUGESal**	13.8882	0.3668	0.4746	0.1936	66.5575	0.9853 / 13.0369 / 25.2893	0.2102 / 0.0458 / 0.0458
M14 - **Multi-task (Ent + QG)**	14.5276	0.3539	0.5749	0.1831	**60.0294**	0.9853 / **14.1828** / 22.2296	0.1985 / 0.0411 / 0.0411
M15 - **Closed book decoder**	13.4158	0.3675	0.4648	0.1925	68.2858	0.9866 / 12.0588 / 27.3686	0.2074 / 0.0444 / 0.0444
M16 - **SENECA**	13.7676	0.3660	0.5233	0.1966	64.9710	0.9880 / 12.3610 / 16.7640	0.2146 / 0.0303 / 0.0303
M17 - **T5**	**19.3891**	0.3833	**0.7763**	0.2140	**59.5288**	**0.9775 / 14.2002** / 12.9565	**0.1810 / 0.0209 / 0.0209**
M18 - **NeuralTD**	12.9241	0.3783	0.3543	0.2038	74.4033	0.9830 / 10.7768 / **12.4443**	0.2645 / 0.0901 / 0.0901
M19 - **BertSum-abs**	14.9525	0.3649	**0.6240**	0.1876	**60.8893**	**0.9517 / 13.9197 / 12.3254**	**0.1697 / 0.0156 / 0.0156**
M20 - **GPT-2** (supervised)	13.9364	0.3678	0.5787	0.1759	**51.8352**	0.9791 / **15.9839** / 15.4999	0.1875 / 0.0362 / 0.0362
M21 - **UniLM**	**15.5736**	**0.4230**	0.5294	0.2084	67.1960	**0.9685** / 11.5672 / **11.7908**	**0.1722 / 0.0180 / 0.0180**
M22 - **BART**	**17.1005**	**0.4271**	**0.7573**	0.2105	62.2989	**0.9771** / 12.8811 / 15.2999	**0.1627 / 0.0127 / 0.0127**
M23 - **Pegasus** (dynamic mix)	**18.6517**	0.4261	**0.7280**	**0.2131**	64.1348	0.9438 / 13.7208 / 11.6003	0.1855 / 0.0355 / 0.0081
M23 - **Pegasus** (huge news)	**17.8102**	0.3912	**0.6595**	**0.2189**	66.7559	0.9814 / 12.9473 / 14.9850	0.1883 / 0.0251 / 0.0251

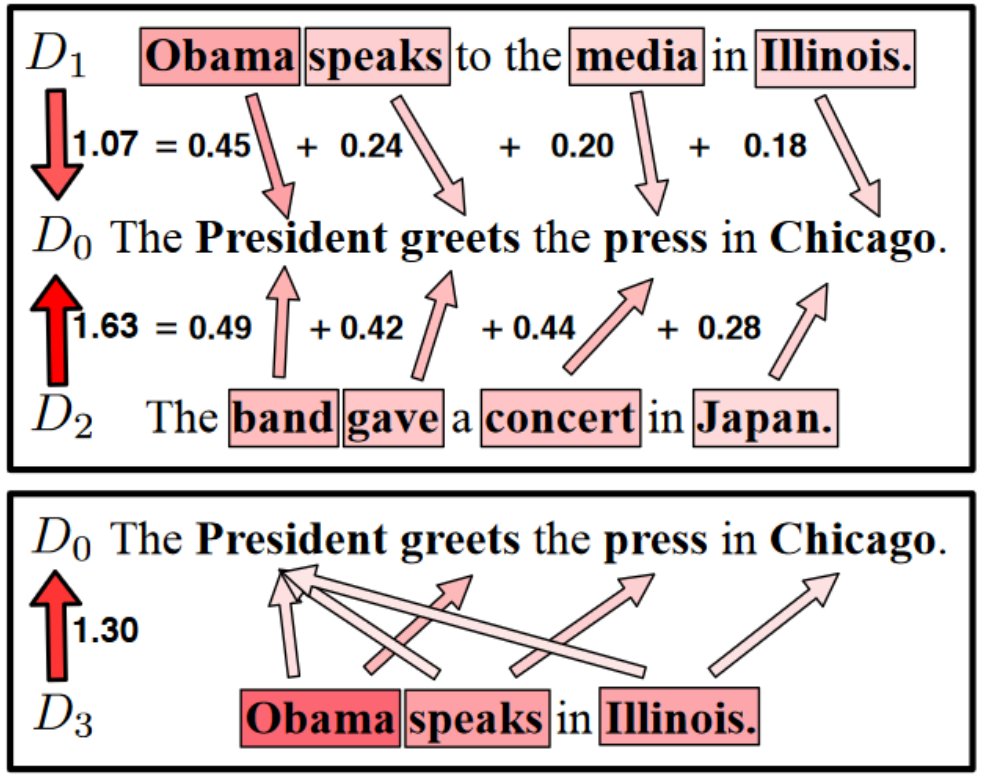
Rouge (2004), Source

ROUGE-N

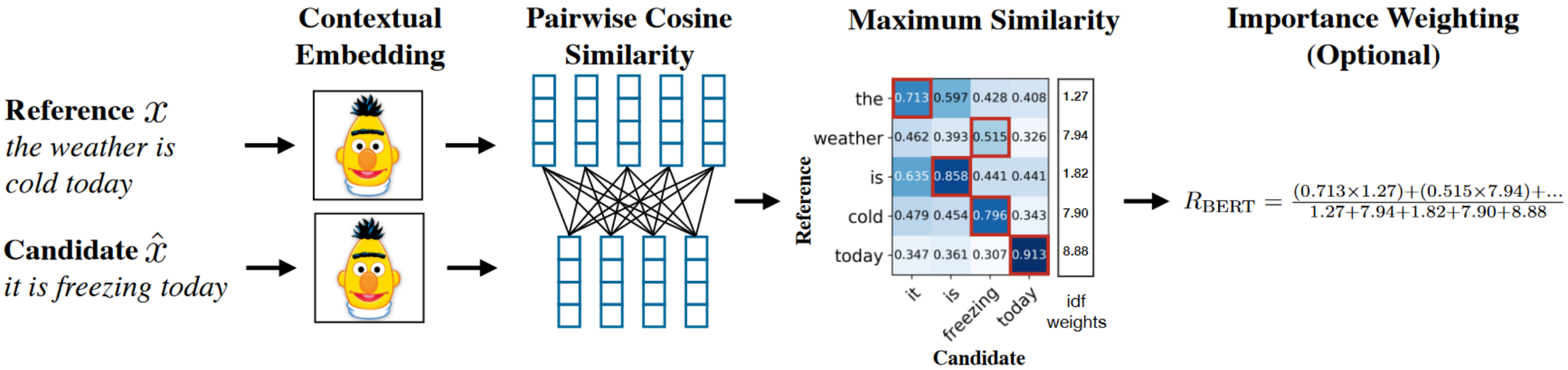
$$= \frac{\sum_{S \in \{ReferemceSummaries\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in \{ReferenceSummaries\}} \sum_{gram_n \in S} Count(gram_n)}$$

WordMover Distance (2015), Source

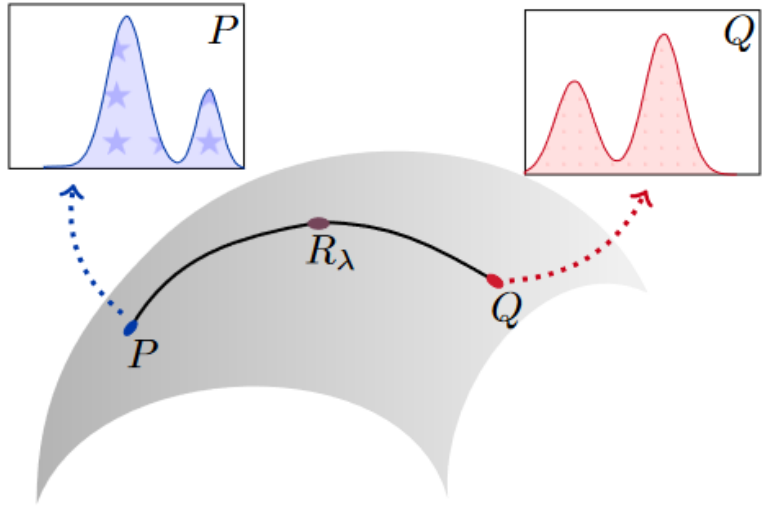


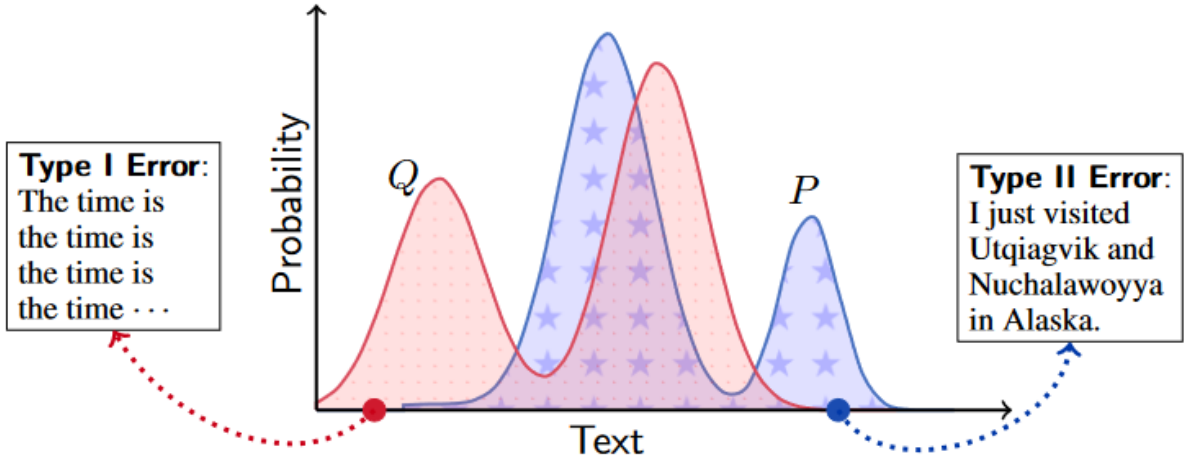


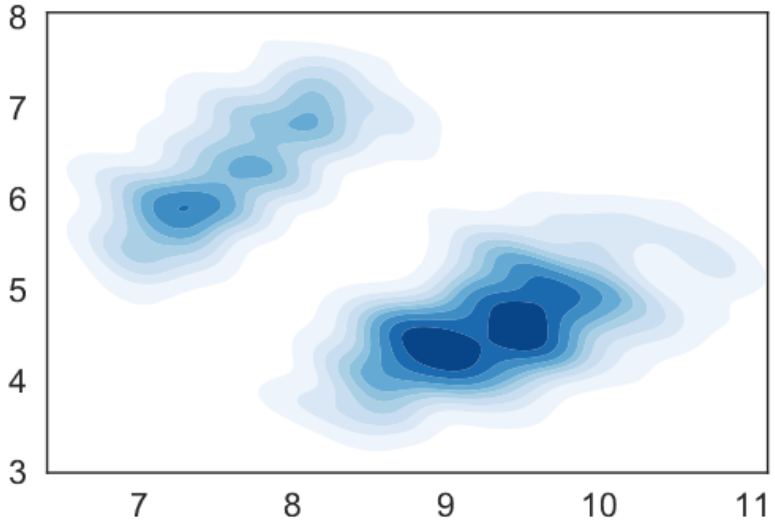
BertScore (2020), Source

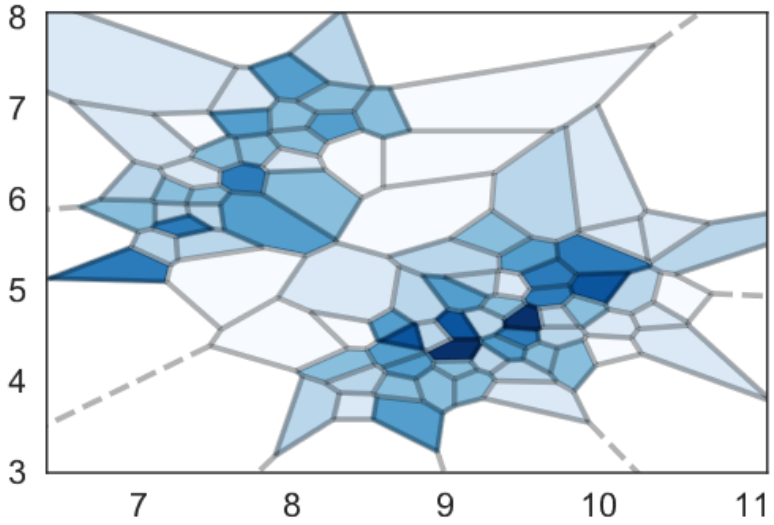


Mauve (2021), Source





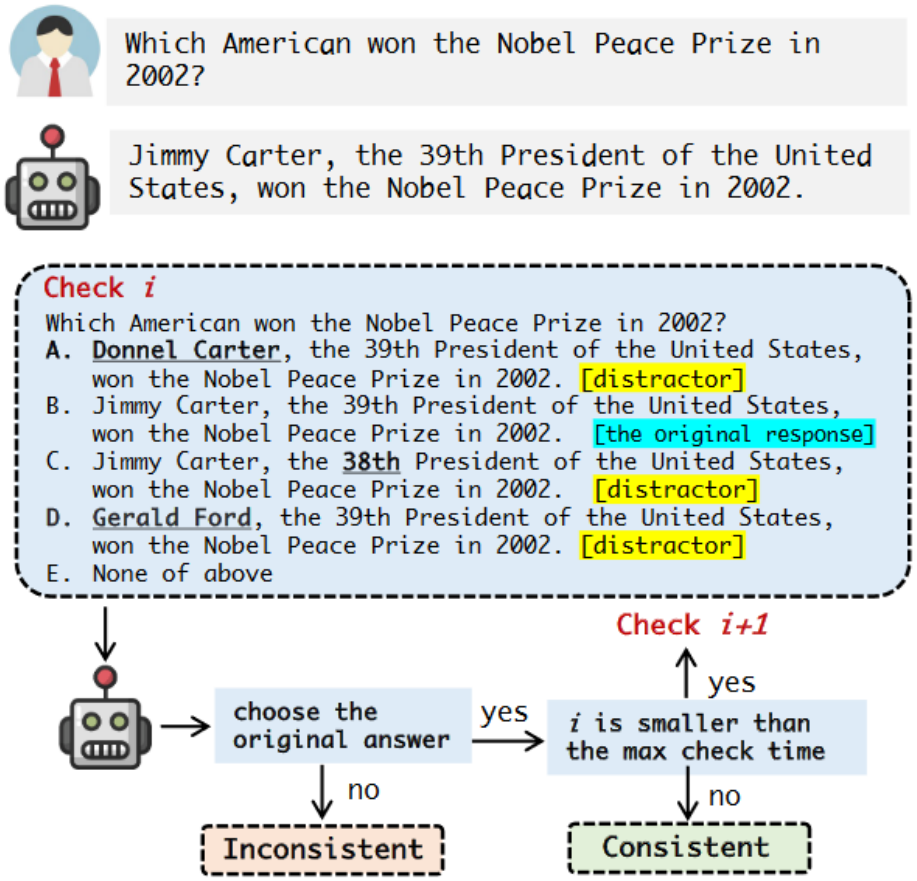




Menli (2023), Source

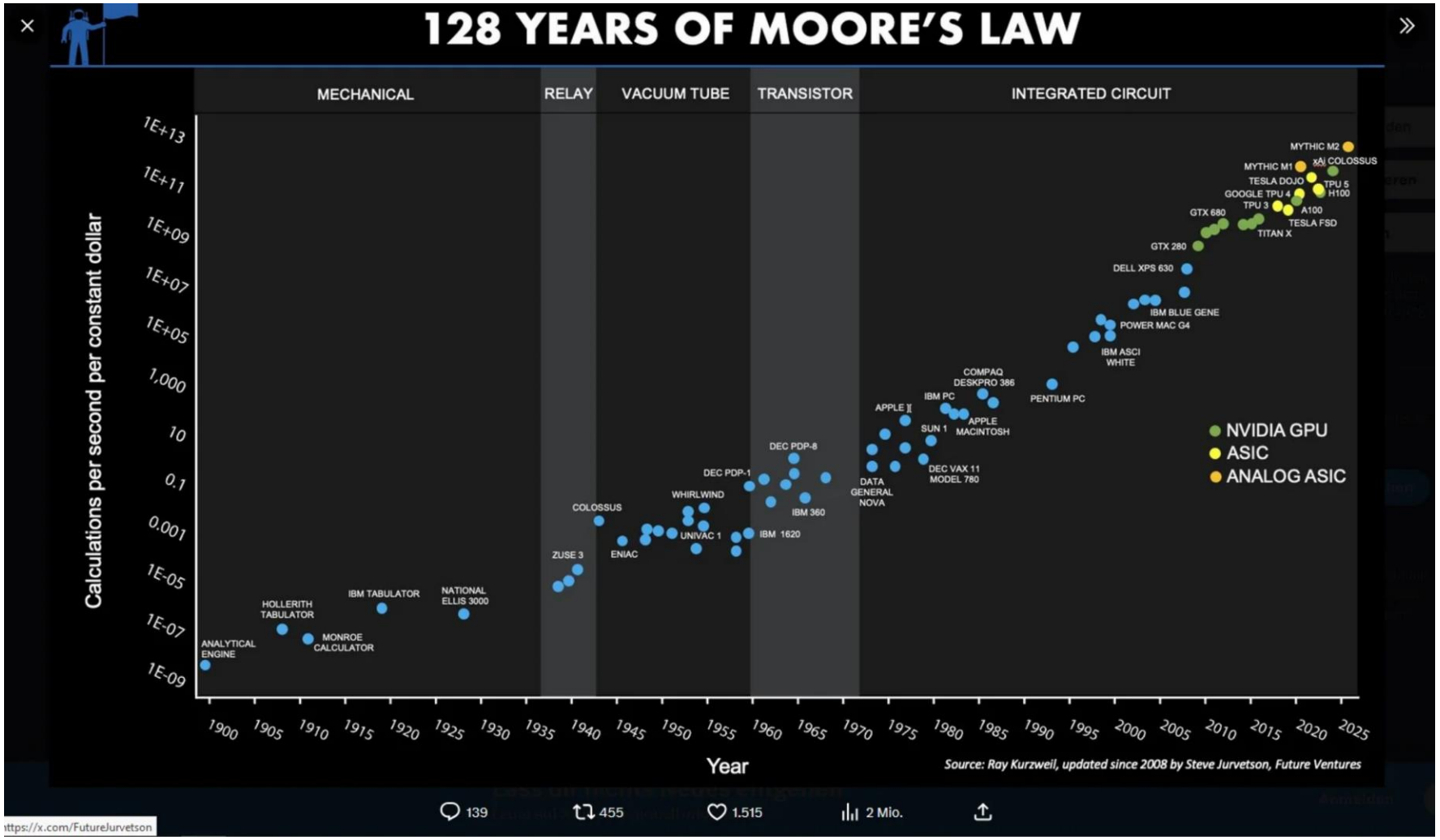
- " Hey LLM are you surprised?"

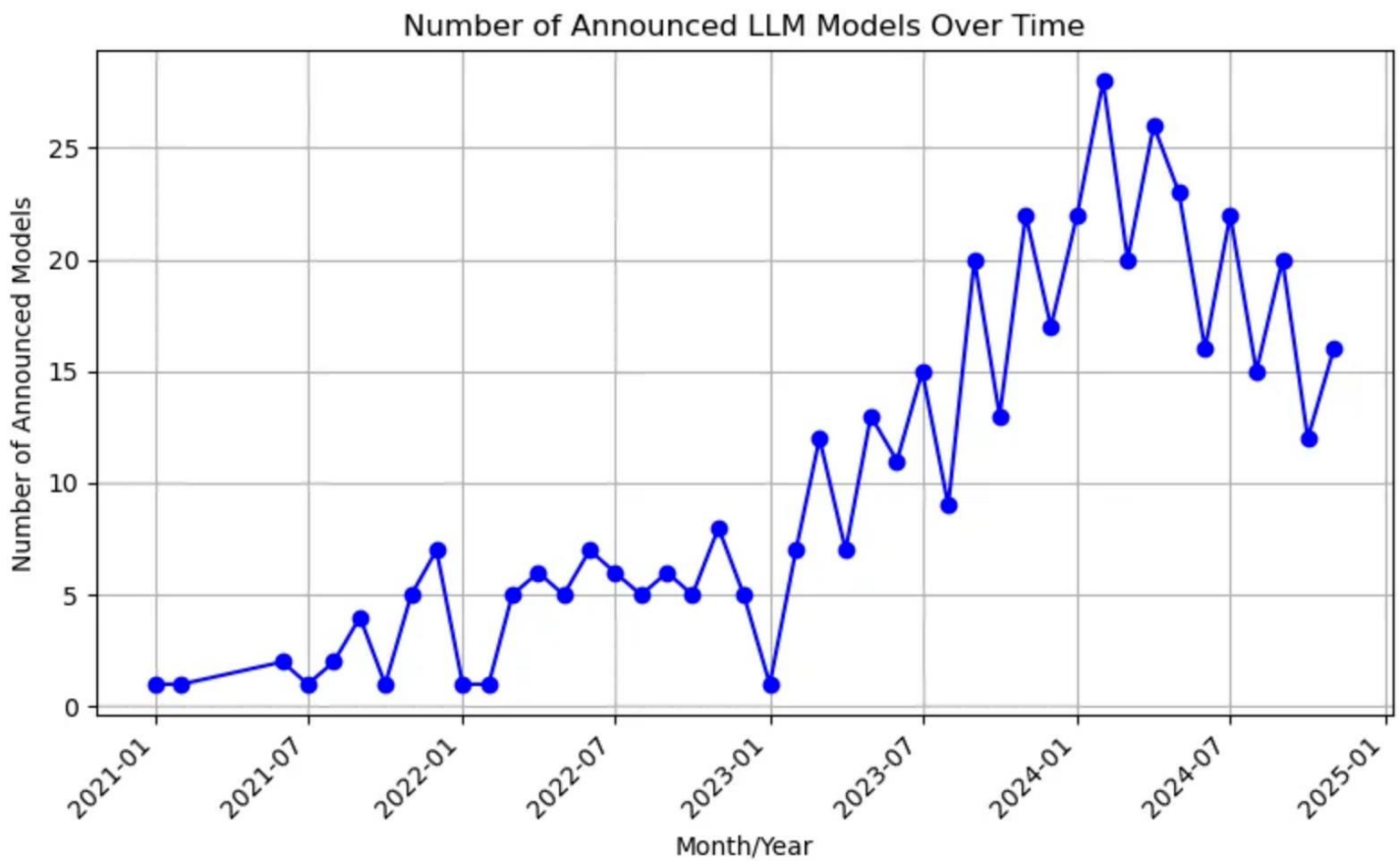
TrustScore (2024), Source



The Bitter Lesson from Rich Sutton, Source

- What's the Difference in time Between static and Dynamic embeddings?




Number of Announced LLM Models Over Time
Number of Announced Models
0
5
10
15
20
25
2021-01
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
Month/Year

Source

The Surprising Effectiveness of Test-Time Training for Abstract Reasoning

Ekin Akyürek Mehul Damani Linlu Qiu Han Guo Yoon Kim Jacob Andreas

Massachusetts Institute of Technology